

IoT Security: Security Solutions for the Internet of Things



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- Every Wednesday from 16.00 to 18.00 in (ITZ) SR 004, from 15.04.2026 until and including 15.07.2026.
- Attendance is advised.
- If the attendees cannot fit in the current classroom, we will try to find a solution.
- Very few slides to learn by heart
- The course (and the exam) is mostly based on **critical thinking** and **basic knowledge** (provided in a few important slides) regarding security and privacy

The following topics will be covered:

- Basic concepts (definition of the Internet of Things (IoT), examination of the different segments of the IoT and their diversity, applications of the IoT, the need for (lightweight) security in the IoT)
- Requirements, specifications, threat modelling, attack surface analysis for the IoT
- The ENISA reports regarding good practices for security of the IoT
- The Industrial Internet of Things (IIoT) and the concept of IoT 2.0

The following topics will be covered:

- Introduction of security in the IoT in the design phase
- Lightweight security primitives: Physical Unclonable Functions (PUFs), True Random-Number Generators (TRNGs), Trusted Platform Modules (TPMs), and other solutions
- Lightweight cryptography for the IoT: Standards and proposed solutions
- Authentication protocols in the context of the IoT
- Attestation of IoT devices in the field
- Advanced security solutions for the IoT utilizing the blockchain, machine learning, post-CMOS technologies and/or advanced artificial intelligence (Security for the IoT 2.0 concept)

The following topics might be covered:

- Secure sensor data aggregation in the context of the IoT: Issues and potential solutions
- Sensor measurements with a Commercial Off-The-Shelf (COTS) evaluation board
- Security in the framework of COTS hardware
- Exploring a comprehensive security solution for a COTS evaluation board in the framework of a network forming a small segment of the IoT

The Internet of Things (IoT): A Definition

- Not a single definition for the IoT
- IBM (<https://www.ibm.com/topics/internet-of-things>): “The Internet of Things (IoT) refers to a network of physical devices, vehicles, appliances, and other physical objects that are embedded with sensors, software, and network connectivity, allowing them to collect and share data.”
- Wikipedia (https://en.wikipedia.org/wiki/Internet_of_things): “The Internet of things (IoT) describes devices with sensors, processing ability, software and other technologies that connect and exchange data with other devices and systems over the Internet or other communications networks.”
- Own (https://tuprints.ulb.tu-darmstadt.de/21494/2/Dissertation_Anagnostopoulos_2022_two-sided.pdf): “The Internet of Things (IoT) refers to a network of devices where data are exchanged, processed, and utilised by the different sensors, actuators, and other electronic devices connected to this network, potentially also leading into actions being taken, without direct human intervention, supervision, or control. Within this framework, devices communicate with each other, and decide on different actions that should be performed by the network’s actuators, based on the data received from the various sensors of the network, as well as from other electronics, and according to a set of predefined rules. The IoT has found wide application both in space and in terrestrial applications, and especially in the implementation of such concepts as smart homes, smart cities, smart grids, and smart vehicles.”

- The Internet of Things
 - The overall interconnected network of “smart” devices
 - The Internet of autonomously operating devices
 - Essentially, the overall network of sensors, agents, actuators, and other devices that perform some task
 - Allows devices to collect information and data, to process them, and to take actions
 - Without human intervention/action
 - Based on sensors that gather information and other data as well as on actuators and agents that perform tasks
 - Includes processing devices that have the ability to decide on actions to be taken based on the data and sets of (potentially predefined) rules
 - Potentially combined with Artificial Intelligence (AI)

- Components of the Internet of Things
 - Sensors and data aggregators: Resource-constrained devices, such as single-board computers and microprocessors
 - Actuators and agents: Potentially resource-constrained devices or ones focusing on a single task, which may be based on commercial off-the-shelf hardware or dedicated hardware with limited functionality
 - Processing segment: Most probably high-end devices, e.g., infrastructure servers
- Segments of the Internet of Things
 - IBM (<https://www.ibm.com/topics/internet-of-things>): physical devices, vehicles, appliances, and other physical objects that are embedded with sensors, software, and network connectivity
 - Wikipedia (https://en.wikipedia.org/wiki/Internet_of_things): devices with sensors, processing ability, software and other technologies that connect and exchange data with other devices and systems over the Internet or other communications networks
 - Own (https://tuprints.ulb.tu-darmstadt.de/21494/2/Dissertation_Anagnostopoulos_2022_two-sided.pdf): different sensors, actuators, and other electronic devices connected to this network / space and terrestrial applications, and especially smart homes, smart cities, smart grids, and smart vehicles
 - World Forum on the Internet of Things (<https://wfiot2023.iot.ieee.org/about>): Space, Maritime, Agriculture & Aquaculture, Smart Cities, Energy, Power, and Sustainability, Industry and Manufacturing
- **Huge diversity of devices, systems, protocols, and networks!**

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- Unlike the Internet, where some standardization has been achieved over the years, the IoT is still based upon a collection of heterogeneous devices and diverse systems that utilize different protocols and networks
- Some examples of connectivity solutions used in the network:
 - Wi-Fi
 - LoRaWAN
 - Bluetooth
 - Ethernet
 - Serial
 - Analog I/O Pins
 - CAN
 - ZigBee
- Problems of incompatibility and lack of standardization...
- Additionally, the relevant devices may range from sensors and evaluation boards to high-end dedicated servers



https://www.eetasia.com/wp-content/uploads/sites/2/2022/05/iot-devices_cover.jpg?w=600&h=400&crop=1

The Internet of Things (IoT): Urgent Need for Security

- **Huge diversity of devices, systems, protocols, and networks!**
 - As well as incompatibilities and lack of standardisation
- **Enormous scale: ~30 billions of connected devices by 2030, ~17 billions today!** (<https://www.statista.com/statistics/1183457/iot-connected-devices-worldwide/>)
- **At the same time, the IoT's operation is by design not being supervised by humans!**

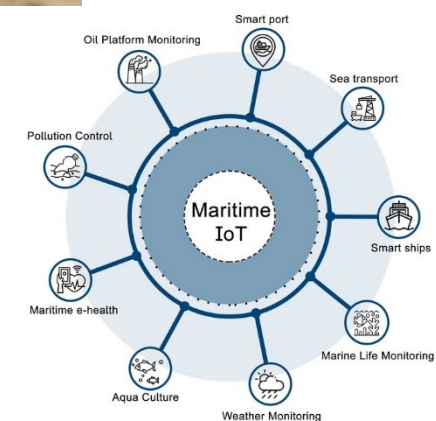


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https://www.researchgate.net/profile/Abderrahmen-Trichili/publication/360233150/figure/fig5/AS:1149675222646785@1651115341591/IoT-categories-and-uses-in-maritime-communications_W640.jpg



The Internet of Things (IoT): Urgent Need for Security

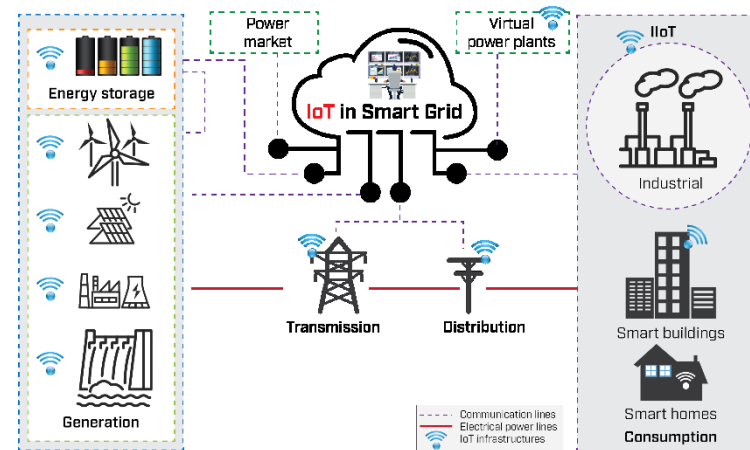
- **Unconventional scenarios, previously unthought of!**
 - One's smart autonomous car automatically opens one's smart garage's door for the car to be parked inside one's garage.
 - So, now, if someone steals one's smart car, do they also gain access to one's home?
 - Ease of use / improvements to user experience (which may lead to previously unforeseen/unintended applications)

- **Connection to critical infrastructures**

- Space segment
- Smart (energy) grids

- **IoT 2.0**

- Connection to 5G/6G, machine learning and AI, edge computing, Industry 4.0, and the blockchain

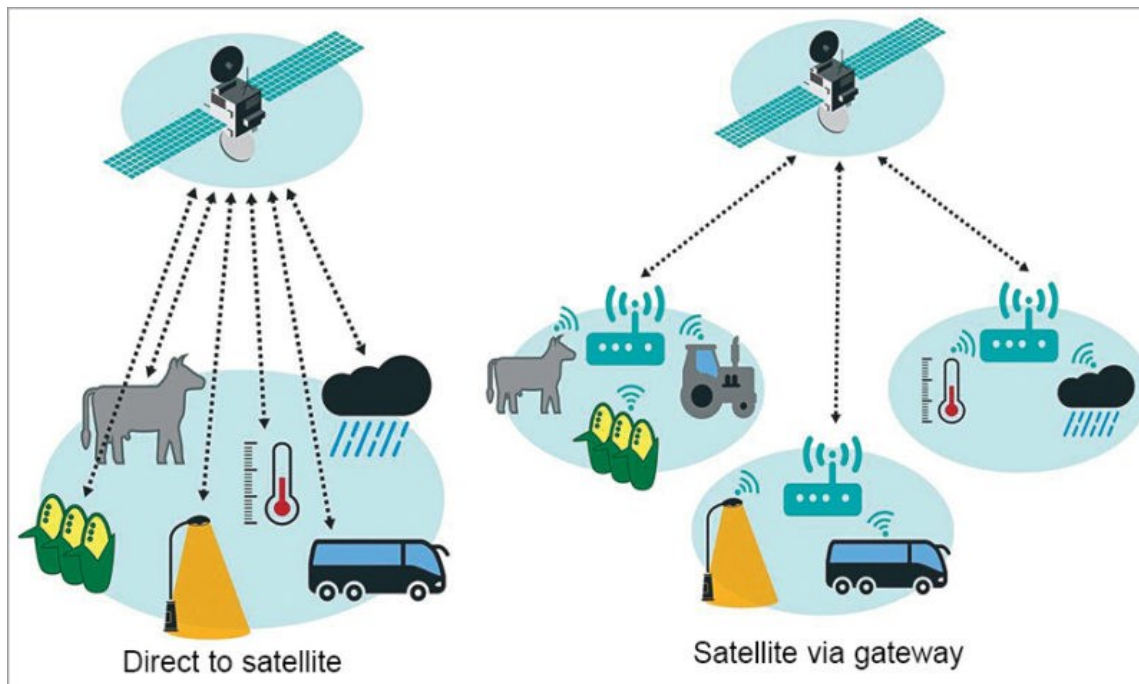


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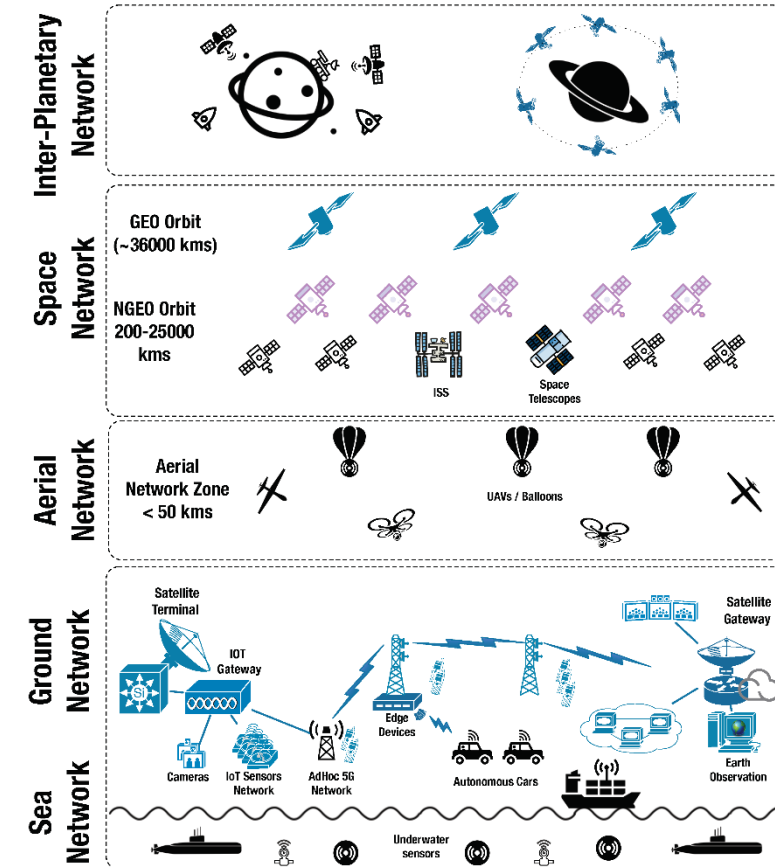
- **Deployment in the open field**
 - Contrary to normal devices, IoT devices may be deployed in the open field and be totally unsupervised
- **Very early standardization effort and mostly regarding network protocols**
 - 5G / 6G as the technology encompassing all others for wireless communication; however LoRa is also getting more and more popular
 - Ethernet for wired communication
- **Relation/balance between security and cost (especially in unforeseen/unintended applications of already deployed devices)**
 - Security and its cost
 - Security and the cost associated with damages
 - Security and ease of use / improvements to user experience (which may lead to previously unforeseen/unintended applications)
 - Security and risk assessment/management (and also ease of use as well as cost/price calculations)
 - Acceptable level of security

The Internet of Things (IoT): Urgent Need for Security

- Essentially, a need to secure systems of systems of increasing complexity as well as vast networks of networks



<https://iot.electronicsforu.com/wp-content/uploads/2020/10/3-696x413.jpg>



<https://www.mdpi.com/1424-8220/21/23/8117>